

Cell Cycle: Copy, Check, Divide

LAUNCH • Science • 40 minutes

I can describe the cell cycle, distinguishing interphase from mitosis and cytokinesis, and sequence prophase, metaphase, anaphase and telophase using chromosome evidence.

Prepare before pupils arrive

Setup: Prepare one clear canonical sequence plus several deliberately ambiguous images. Keep interphase outside the mitosis-stage strip while showing that both belong to the wider cell cycle.

Safety: Use prepared slides or approved images only. Do not imply that every real cell fits a perfectly discrete stage or that chromosome drawings show molecular scale.

Equipment: Stage cards, chromosome-copy counters, prepared diagram or micrograph set, sequencing grid and command-word frame.

Safety: This is a model-and-image lesson with no live tissue collection. If prepared slides and microscopes are used, carry microscopes with two hands, keep slides flat and report damaged glass; use only teacher-approved images.

Fallback: Use enlarged tactile cards, spoken choices or eye-pointing while an adult records the pupil's exact sequence and reasons. The same formal vocabulary and evidence standard apply.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Edexcel GCSE Biology 1BI0 Foundation • Paper 1 • Topic 2 • specification point 2.1 • the cell cycle and mitosis.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival • 4 min	State where chromosomes are found in an animal or plant cell. Explain why DNA must be copied before one cell forms two cells.	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today's answer.
Starter • 4 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask 'What makes you think that?' and preserve the prediction for later evidence.

Stage / total time	Teaching move	Adult support / response
I do · 5 min	interphase — prepare and copy mitosis — divide the nucleus cytokinesis — split the cell Begin with one parent cell. During interphase it grows, increases subcellular structures and replicates its DNA so each chromosome has a copied partner.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.
We do · 5 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 4 min	Sequence the evidence: prophase — chromosomes condense; metaphase — chromosomes align near the cell equator; anaphase — sister chromatids separate towards opposite poles. Complete the sequence: telophase — two chromosome groups reach the poles and new nuclei form; cytokinesis then divides the cytoplasm and membrane.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 4 min	Route each event to preparation before nuclear division, nuclear division, or cell split/outcome, then build one defensible sequence. Read one event card and identify what changes.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 10 min	Create an evidence-labelled cell-cycle sequence and answer GCSE DESCRIBE and EXPLAIN questions that distinguish interphase, mitosis and cytokinesis. Record: A complete ordered cycle, at least four stage-evidence links, a correct distinction between DNA replication and chromosome separation, and one pupil-owned explanation.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	Describe the cell cycle from interphase to two daughter cells using at least four formal stage terms. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Give one next move; the pupil edits or re-attempts on the existing work.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: The cell replicates its DNA before nuclear division.

Correct. DNA replication occurs during interphase; chromosome arrangement and separation occur during mitosis.

Misconception repair

Repaired. Replication prepares two DNA copies; mitosis distributes them, so chromosome number is maintained in each daughter cell.

Guided checkpoint

Correct. DNA replication occurs during interphase; chromosome arrangement and separation occur during mitosis. Ask which event prepares the genetic material before nuclear division begins.

Cell growth

INTERPHASE · PREPARE. Interphase: the cell prepares the material and structures needed for division. This happens before chromosome copies are separated.

DNA replication

INTERPHASE · PREPARE. Interphase: genetic material is replicated before mitosis. DNA copying prepares for mitosis; it is not cytokinesis.

Chromosomes condense

MITOSIS · NUCLEAR DIVISION. Mitosis: chromosome condensation is evidence of prophase. This is a change in chromosome appearance during nuclear division.

Chromosomes align

MITOSIS · NUCLEAR DIVISION. Mitosis: alignment positions chromosome copies for separation. The chromosome copies have not yet formed two cells.

Sister chromatids separate

MITOSIS · NUCLEAR DIVISION. Mitosis: chromosome copies are separated into two sets. This is nuclear-division evidence, not DNA replication.

Two nuclei reform

MITOSIS · NUCLEAR DIVISION. Mitosis: two daughter nuclei form around equivalent chromosome sets. The nuclei reform before or while the cell finishes splitting.

Cytoplasm divides

CYTOKINESIS · CELL SPLIT. Cytokinesis: physical cell division completes the cycle. This card describes the cell split, not chromosome copying or separation.

Model limitation

Stage cards freeze a continuous, three-dimensional process into neat snapshots. Chromosomes, spindle fibres, cell size and timing are not to scale, and checkpoints and molecular controls are mostly omitted.

Independent evidence

A complete ordered cycle, at least four stage-evidence links, a correct distinction between DNA replication and chromosome separation, and one pupil-owned explanation.

Supported: Order six labelled picture cards, match prepare/divide/split, and complete a stem explaining when DNA is copied. Use: During interphase ___. During mitosis ___. During cytokinesis ___.

Standard: Sequence interphase, four named mitotic stages and cytokinesis, annotate visible evidence, then explain why DNA replication must precede mitosis. Use DESCRIBE for the ordered changes and EXPLAIN for cause plus biological reason.

Stretch: Evaluate an incorrect cell-cycle model, repair chromosome-number reasoning and explain why stage boundaries can be uncertain in a snapshot. Distinguish observation, stage inference and model limitation; include the caveat that mutation or segregation error can disrupt the usual outcome.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre's authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner's practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: SCI_L_W9L1_Cell_Cycle_Introduce.html

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.